

Concept Note

on

**“Towards Socio-Economic Empowerment of small and marginal
Farmers of Pune district through Sustainable Agriculture
Practices”**

Submitted by:

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1. Executive Summary

Particulars	Description
Project Goal	The overarching goal of the project is to develop climate adaptive and resilient livelihood system through a package of sustainable agricultural practices, technological interventions
Geography	The project would be implemented in selected villages of Pune district of Maharashtra
Project Objectives	● To ensure secure livelihoods and prosperity of 500 farmers through inclusive, efficient, resilient, climate smart and scientifically recommended agricultural practices that would not only ensure their food and nutritional security but at the same time safeguard the environment as well ● To prevent distress migration from rural to urban areas, due to lack of remunerative employment opportunities for a decent standard of living ● To reduce the drudgery, facilitate timeliness in farm operations and efficient use of inputs through farm mechanization ● To establish and nurture village level community-based institutions for sustaining the project gains in the post project period
Target Beneficiaries	The target beneficiaries are 500 small and marginal farmers from the selected villages
Project Duration	24 Months
Project Budget	Rs. 85 Lakhs

2. Rationale of the Project

Climate change remains the biggest threat to Indian agricultural system as it can wreak havoc on the lives and livelihoods of millions of farmers in the country. The vagaries of climate can disrupt food availability, reduce access to food, and affect nutritional aspects of food. The state of Maharashtra is highly acclaimed for its development model which has exhibited phenomenal growth in social, economic and environmental indicators. However, this remarkable achievement is marred by the fact that the state's agricultural performance is declining at an alarming rate especially during the last few decades.

The agriculture sector of the state is plagued with various challenges like poor yield, small and fragmented landholdings, traditional methods of agriculture and harvest and post-harvest losses. Moreover, the agriculture is mainly rainfed and owing to semi-arid region and erratic rainfall, the productivity of the crops is much lower than its actual potential.

Some of the major agricultural issues in Pune district of Maharashtra are enumerated as below:

- **Lack of knowledge on modern methods of cultivation:** Farmers are primarily practicing traditional methods of agriculture and hence agriculture is characterized by a low level of quality inputs, lack of climate-smart agricultural practices, and poor management practices.
- **Small and Marginal Farmers:** In Maharashtra, more than 78.4%¹ of the farmers belong to the category of small and marginal farmers. Climate change pose a significant threat to small and marginal farmers as they depend on rain-fed agriculture, cultivate marginal areas, and lack access to technical or financial support that could help them invest in more climate-resilient agriculture.
- **Acute water shortage:** The region is historically known as drought prone region, however the frequency and intensity of droughts have increased over the past decades. The region is reeling under acute water crisis due to repeated failures of monsoon and climate change
- **Distress migration:** Migration is predominant in the district especially after the harvesting of Kharif crops. Due to scarcity of water, farmers don't take crops during rabi season and hence migrate to other cities searching for remunerative livelihood opportunities.

3. Proposed project components

“The future of India lies in its villages”. This famous statement by Mahatma Gandhi stands true even when the world is hurrying towards industrialization and urbanization. Indian villages are brimming with untapped potential which needs to be harnessed by building an ecosystem of sustainable rural livelihoods. The project focuses on building resilience of small and marginal farmers through climate resilient agricultural practices.

Sustainable Agriculture Practices

The Sustainable agriculture can be defined as ecologically friendly farming practices that are economically viable, environmentally sound, and protect human health. It enables society to meet their food and textile needs without compromising the needs of future generations. Hence, Climate Resilient Agriculture (CRA) practices will be undertaken for sustainable agricultural development and ensuring food security while mitigating the negative impacts of climatic changes on agricultural productivity. CRA practices will be promoted to achieve higher output, yield, and reduce greenhouse gas emissions. This will be achieved by the adoption of the following steps/ processes:

1) Soil Testing:

Indiscriminate usage of chemical fertilizers not only hampers the productivity of crops but also leads to deterioration of soil structure, wastage of nutrients, destruction of soil micro-organisms, scorching of plants in extreme cases and an unnecessary escalation in cost of cultivation. Judicious use of fertilizers based on soil testing will be promoted both for maintaining soil productivity and reducing the input cost of crop cultivation.

Activity Description:

¹ <https://www.adb.org/sites/default/files/linked-documents/53264-001-ssa-01.pdf>

Soil testing will be carried out with an envisaged objective of educating farmers about soil productivity and fertility. Soil samples from the farmer's field shall be collected and sent to one of the renowned laboratories for the analysis of macro and micro nutrients. Soil test shall help in identifying the crop based nutritional requirement. Farmers would subsequently then apply the recommended doses of nutrients as per the information indicated in the soil health card and realize the benefit of crop productivity.

Expected Outcome:

- Awareness on judicious use of fertilizers for maintaining soil health
- Improvement in crop production and productivity
- Reduction in cost of cultivation

2) Adoption of organic methods:

The presence of adequate organic material in the soil in the form of humus is very important for improving the fertility of soil and uptake of nutrients. Farm compost is an organic fertilizer that can be made on the farm from crop residues or animal manures at a very nominal cost. This is a low-cost yet effective method for improving the productivity of crops. In addition to farm compost, other organic practices like bio-fertilizers, green manure, crop rotation, mulching etc will also be introduced. This will not only reduce the cultivation cost but would also repair, conserve and enhance the ecological balance.

Activity Description:

Demonstrations on organic methods will be given to the farmers.

Expected Outcome:

- Significant improvement in soil health
- Reduction in the use of chemical insecticides and fertilizers through the adoption of effective eco-friendly practices.
- Reduction in cost of cultivation

3) Adoption of System of Rice Intensification:

The System of rice Intensification(SRI) is a climate smart farming methodology which aims to increase the productivity of rice by changing the management of soil, plants, nutrients and water. SRI is a low-water, **labour intensive method** of rice cultivation which promote less reliance on artificial fertilizers, pesticides, herbicides, and other agrochemicals while reducing the emissions of greenhouse gases.

Activity Description:

System of rice Intensification would be promoted for enhancing the productivity of paddy while improving soil health and fertility.

Expected Outcome

- Enhance production of rice and income of farmers
- Strengthens the resilience of crops to climate change

4) Crop Diversification

Indian agriculture has taken giant strides in the last few decades where it has transitioned from traditional cropping system to high value cultivation of crops. Indian economy too has witnessed shifting of consumption pattern from traditional cereals to high value agricultural commodities. Hence, it is proposed to diversify the sources of income of farmers through cultivation of staples, vegetables and cash crops on small plot during khariff and rabi season. Also, dedicated efforts will be taken to promote millet cultivation as they are known to provide food and nutritional security to the millions of households in Indis.

Depending upon the localized agroclimatic conditions, suitable crops will be identified and support would given to farmers in the form of improved seeds, fertilizers and technical expertise.

Expected Outcomes

- Crop diversification to develop a portfolio of income sources
- Food and Nutritional security

5) Integrated Crop Management:

Continuous use of suboptimal doses of nutrients in the intensive cropping system has led to severe depletion of nutrient reserves in the soil, causing multiple nutrient deficiencies, deterioration in soil physical properties and biological activity, water, and ecosystem, thereby decline in crop productivity. Sustainable crop rotation, along with use of organic manures, crop residues, bio-fertilizer, and fertilizers in an integrated manner helps to restore and sustain soil fertility and crop productivity. It also helps to check the emerging deficiency of micro-nutrients **other than** NPK (Nitrogen, Phosphorus and Potassium). Integrated Pest Management is an eco-friendly approach that encompasses cultural, mechanical, and biological, and need-based chemical control measures that help to reduce input cost for crop production and maintain water pollution, ecological balance in agriculture.

Activity Description:

Promotion of climate smart agriculture practices like seed treatment with bio fertilizer, crop rotation, integrated nutrient management and integrated pest management practices will be promoted.

Expected Outcome

- Better yield due to balanced nutrition to the crops
- Boost Soil fertility

6) Promotion of kitchen gardening:

Kitchen gardens can be established and maintained on a small patch of land with minimum technical inputs. These gardens provide the rural resource to the poor communities with a platform for innovations in supplemental food production as well as an opportunity to improve their livelihoods. Kitchen gardens are cost-effective, practical, and can easily meet the balanced dietary requirements of rural households as well as add substantially to the family income.

Activity Description:

Kitchen gardens will be promoted for ensuring availability of green and leafy vegetables for home consumption. The nutritional garden will have both perennial and seasonal vegetables and medicinal plants.

Expected Outcome

- Nutritional Security

7) Farm Mechanization

Farm mechanization enhances production, productivity and profitability in agriculture by achieving timeliness in farm operations. Small and marginal farmers are not able to leverage the full potential of farm mechanization due to constraints like hilly topography, socio-economic conditions, high cost of transport etc.

Activity Description:

A custom hiring centre would be established in the village which would be stocked with different farm machineries like rotavator, zero till drill, drum seeder, multi-crop planter, power weeder, chaff cutter, tractor etc. The beneficiaries will also be trained to operate these machineries and use in their respective farm fields.

Expected Outcome

- Provides access of costly farm machinery to social and marginalized farmers especially female farmers
- Reduces drudgery
- Facilitates timeliness in farm operations and efficient use of inputs

8) Promotion of micro irrigation

Micro irrigation is a scientific method of irrigation carrying desired water and nutrients directly to the root zone of the plant on a drop by drop basis. Drip or trickle irrigation is a type of micro irrigation system that efficiently delivers water and nutrients directly to the plant's root zones in the right amounts which subsequently help the plants to grow in an optimal manner. Hence, a financial assistance would be given to the selected farmers for the adoption of micro irrigation system which would not only enhance the production and productivity of crops but would also save water requirements.

Expected Outcome

- Water requirements will be saved by 40%.

9) Establishment of Village Knowledge Center:

Processes will be initiated to evolve, establish and nurture knowledge centre at the village level to ensure convergence, harnessing social capital and effective utilization of existing agricultural system. Knowledge center will act as a nodal point to interact with scientists and officials at one end and with progressive farmers at the other end. It would ensure a pool of local (progressive farmers) as well as scientific knowledge (State Agricultural Universities and Krishi Vigyan Kendra). One of the most important activity of knowledge center, will be, to ensure convergence of existing systems with crop production and effective implementation of government schemes like MGNREGA (Mahatma Gandhi National Rural Employment Guarantee Act), irrigation etc. It would also become node for regular interactions at the village level and support actions at the local level. These will be manned by the local villagers (youth farmers) with the help of social capital (retired officials, progressive farmers and locally available experts).

Expected Outcome

- Linkages with various government departments for leveraging existing schemes.

10) Newsletter Development:

Regular dissemination of right information to the right persons in a right mode is key to any successful intervention. Farmers are not getting information on a regular basis and in a format which suits to their need. Indian extension system's outreach is very scanty in villages and most of the extension posts are lying vacant in rural India. Few resourceful people get information and that too, not on a regular basis. Social and marginalized farmers especially Women farmers are left out of this process. Hence, to fill this gap and disseminate information, a quarterly newsletter named Kisan Samvad will be brought out in local languages which will be outcome of regular interaction between scientists, experts, writers and field activists. Regular workshops will be held to bring out this newsletter and it will be disseminated to these farmers. Emphasis will be given to identify actionable interventions in each month's newsletter so as to bring desirable results.

Expected Outcome

- Dissemination of Credible Information to enhance crop yield

4.Sustainability**Formation of Farmer Interest Groups:**

Village Institutions are a very important part of rural development, they bring out the real sense of democracy and helps in the implementation of a project with increased village level participation and village representation. Farmers Interest Groups (FIG) is an independent and self-managed group of 15 to 20 farmers in cluster farmlands with a shared goal and interest, who work together with their existing resources to achieve the desired goal in selected project villages.

Community Resource Persons:

In each village, a female and a male person will be developed for implementing project activities along with the project staff. During the project period, they would support project team in developing rapport with the community, facilitate implementation of village level activities and will assist in the implementation of various activities promoted by the project in the post project period.

5.Result based monitoring framework

Result based monitoring and evaluation framework would be adopted to track and assess the performance of the project and in the end measure impact through continuous evaluation

Objectives	Activities	Outputs	Outcomes	Impact
1.To ensure secure livelihoods and prosperity of 500 farmers through inclusive, efficient, resilient, climate smart and scientifically recommended agricultural practices that would not only ensure their food and nutritional security, at the same time safeguard the environment as well 2. To prevent distress migration from rural to urban areas due to lack of remunerative employment opportunities for a decent standard of living 3.To reduce the drudgery, facilitate timeliness in farm operations and efficient use of inputs through farm mechanization 4.To establish and nurture village level community-based institutions for sustaining the project gains in the post project period period	1. Formation of Farmer interest groups 2.Soil testing 3.Adoption of organic methods 4.Adoption of system of rice intensification 5.Crop Diversification 6.Integrated Crop Management 7.Promotion of Kitchen Gardening 8.Farm Mechanization 9.Promotion of micro-irrigation 10.Establishment of Village Knowledge Center 11.Newsletter Development	1.500 Families would be trained in improved agricultural practices. 2.200% cropping intensity 3.CRPs would be identified and trained	1. Increase in area under vegetable cultivation. 2. Cultivation of High value Crops. 3. Food and Nutritional security 4. Trained farmers will cultivate crops throughout the year. 5. Cash flow throughout the year Development of a Cadre of Community Workers	1.Farmers income will be enhanced by 65%. 2.Enhanced quality of life. 3.Reduction in distress migration 4.Reduction in agri inputs expenses 5.Reduced emissions of greenhouse gases Sustainability of the project initiatives